



Could enterprise resource planning create a competitive advantage for small businesses?

Competitive
advantage for
small businesses

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Abstract

Purpose – This paper seeks to show how enterprise resource planning (ERP) could create a competitive advantage for small- and medium-sized enterprises (SMEs).

Design/methodology/approach – The main methods used in this study were questionnaires and interviews based on the application of an uncertainty diagnosing business model. Data were collected, using a questionnaire administered to 126 SMEs, in the form of percentage contributions of the underlying causes of uncertainty (structured in the business model) on product late delivery. Analysis of Variance (ANOVA) was carried out in SPSS to analyse the effects of the underlying causes of uncertainty in SMEs.

Findings – ERP could create a competitive advantage in delivery for SMEs by being responsive and agile to change, but not to uncertainty. Results suggested that only a few features in an ERP system were used to deal with change due to uncertainty. It was found that SMEs generally use their ERP system to generate a plan for production and use it as a guideline. SMEs concurrently use a range of buffering or dampening techniques to tackle uncertainty for creating a competitive advantage in delivery.

Research limitations/implications – The application of the business model in SMEs has provided useful knowledge to make-to-stock (MTS), make-to-order (MTO) and mixed-mode (MM) manufacturing enterprises in which underlying causes of uncertainty were significantly affecting their product late delivery performance.

Originality/value – This is a highly original application of an uncertainty diagnosing business model to SMEs using ERP systems.

Keywords Small to medium-sized enterprises, Resource management, Uncertainty management, Competitive advantage

Paper type Research paper

Introduction

The new economy entails e-business and knowledge-driven enterprises that could lead to more responsive and agile methods to deal with change and uncertainty. Responsiveness and agility are the key competitive advantages for manufacturing enterprises. Agile manufacturing can be defined as the capability to survive in an environment of continuous and unpredictable change by reacting quickly and effectively to changing markets, driven by customer-designed products and services (Gunasekaran, 1998). Manufacturing responsiveness relates to the ability of manufacturing systems to make a rapid and balanced response to the predictable and unpredictable changes that characterise today's manufacturing environments (Gindy, 1998). Uncertainty can be defined as unpredictable events in manufacturing environments that disturb the operations and performance of an enterprise (Koh and Saad, 2002). Some studies have referred to this uncertainty as disturbance (Lindau and Lumsden, 1995; Frizelle *et al.*, 1998; Saad and Gindy, 1998). In this paper, change can be defined as the effect of uncertainty on parameter variations and adjustments made to reflect the current status of a manufacturing system within a



production planning and control system. Manufacturing enterprise customers often demand shorter lead-time, near perfect or even perfect delivery service, reduction of product life cycles and increasing customisation. Hence, there is pressure on managers to rapidly improve their enterprises' performance and adapt to change and uncertainty in order to maintain a competitive advantage.

To compete in the new economy, small- and medium-sized enterprises (SMEs) are now facing challenges to become more responsive and agile. This requires re-evaluation of the viability of the existing production planning and control systems to create a competitive advantage for the enterprise. These systems include material requirements planning (MRP), manufacturing resources planning (MRPII) and enterprise resource planning (ERP). These systems are related in that MRP is at the heart of MRPII systems (Parker, 1996) and ERP systems. MRPII effectively plans all resources of a manufacturing company. It is made up of a variety of functions, each linked together: business planning, sales and operations planning, production planning, master production scheduling, material requirements planning, capacity requirements planning, and the execution of support systems for capacity and materials. Output from these systems is integrated with financial reports such as the business plan, purchase commitment report, shipping budget, and inventory projections in monetary value. MRPII is a direct outgrowth and extension of closed-loop MRP. The term ERP can be defined as an accounting-oriented information system for identifying and planning the enterprise wide resources needed to take, make, ship, and account for customer orders. An ERP system differs from the typical MRPII system in certain technical requirements (Gaither and Frazier, 2002). Using MRP, MRPII or ERP systems as a type of production planning and control tool requires the same MRP release logic (Enns, 2001; Koh and Saad, 2002, 2003a). Therefore, the outputs, i.e. planned order release (POR) schedules generated from such planning tools are identical.

The supply chain competitiveness between supplier and customer relies on how effective and efficient the order and information is being handled between the parties in the supply chain. In the business-to-business (B2B) market, many SMEs now have to be able to provide a level of service that is compatible with their corporate customers. Those SMEs that can provide such service could win the supply contract. To excel in the B2B and the business-to-consumer (B2C) markets, an ERP system plays a significant role in production planning and control. Many large enterprises have already utilised such systems for their B2B and B2C business activities. The big ERP system vendors are SAP, Baan, ORACLE, JDEEDWARDS, and PeopleSoft. The implementation cost of such systems is very high, and thus it is difficult to justify to SMEs the costs and benefits of these systems. To cater for the needs of SMEs, many midrange and less complex systems have been developed, e.g. Alliance Manufacturing (Exact Software), MFG/PRO (QAD), WinMan (TTW) and all-in-one (SAP). In conjunction with using such systems as a planning and control tool, many SMEs combine this with other execution concepts, such as just in time (JIT), optimised production technology (OPT) and advanced production scheduling (APS). Within batch manufacture, the combination with JIT is the most common practice.

Despite the rapid development of the mid-range ERP systems for SMEs, little research can be found in evaluating the extent to which ERP could create a competitive advantage for SMEs, particularly on how change and uncertainty could be managed in

such an environment. It is suggested that if change and uncertainty could be managed and responded to, effectively and efficiently, then SMEs could gain a competitive advantage via improved delivery performance. The problem is how can SMEs manage change and uncertainty using a more holistic approach, considering the diverse range of uncertainties that could disturb production.

Literature review

This problem has driven much research into suitable techniques to manage change and uncertainty. Comprehensive reviews (Yeung *et al.*, 1998; Guide and Srivastava, 2000; Koh *et al.*, 2002) suggest that researchers have used a range of buffering or dampening techniques to tackle uncertainty and Koh *et al.* (2000a) showed that practitioners used similar techniques. Safety stock and safety lead-time were found to be robust to various uncertainties. However, a solution for various types of manufacturing environment in SMEs has not been identified. Recent research (Muscatello *et al.*, 2003) investigated implementation issues in SMEs but did not tackle the issues of change and uncertainty.

Grasso and Taylor (1984) advocated the use of safety stock in dealing with late material delivery. New and Mapes (1984) indicated that application of mean yield rate and fixed buffer stocks is appropriate for continuous schedule, make-to-stock (MTS) environments. Wacker (1985) proposed that safety stock could be used robustly to buffer external supply, external demand and internal supply uncertainties for variations in timing and quantity. Yano and Carlson (1985) advocated the use of safety stock against using re-planning of MPS to maintain customer service, because the cost due to frequent re-planning exceeds the cost of safety stock required to maintain a stable schedule. Vargas and Dear (1991) identified that safety stock should be used robustly to buffer external demand and internal supply uncertainties.

In contrast, Sridharan and Laforge (1989) found that safety stock might not be such a robust technique when the level of uncertainty is high. Using simulation modelling, they examined the impact of safety stock on schedule instability, cost and service. It was concluded that efforts to reduce set-up costs and improve forecast accuracy are useful techniques to buffer or dampen schedule instability. Their results suggested that safety stock should be used with caution if it is introduced for the purpose of stabilising schedules. They reported that more nervousness might be produced when using safety stock. Sridharan and Berry (1990) found that in an uncertain demand environment the order-based freezing method could perform better. In addition, they also found that a longer frozen interval produces higher costs.

Atwater and Chakravorty (1994) identified that the use of safety capacity at non-constraints above the system's constraints capacity to be effective in buffering external supply uncertainty, late supply, internal supply uncertainty, machine breakdown, internal demand uncertainty and quality variation. To tackle uncertainty at external demand and external supply, Ho *et al.* (1995) developed a framework to dampen the system nervousness caused by these uncertainties. The framework suggested the use of safety stock, safety capacity, safety lead-time and rescheduling to tackle these uncertainties. Further, research by Ho and Carter (1996) simulated static dampening, automatic rescheduling and cost-based dampening techniques to tackle external demand uncertainty. The simulation results indicated that the performance of the techniques depends on the operating environment of the manufacturing system.

The results also showed that a reduction in the frequency for rescheduling does not necessarily lead to better system performance. They concluded that the system improvement was dependent on the appropriate use of the dampening techniques and the lot-sizing rules.

Grubbstrom and Molinder (1996) suggested the use of safety production plans to deal with demand uncertainty using a Laplace Transform Methodology. The generation of the safety production plans did consider safety stock and safety lead-time. It was identified that as uncertainty in demand on the highest level of a multi-level system increases over time, it amplifies the uncertainty in the current production schedules at the lower levels. Despite the fact that there might well be time to reschedule the top-level plan before it is realised, they proposed that it was important to capture the effects of cumulative uncertainties in external demand at the highest level. Thus, they suggested that the future cumulative demand uncertainties have to be dampened by using a safety production plan. To deal with forecast inaccuracy Krupp (1997) proposed a statistical model that expresses deviations in units of time rather than quantity to provide safety stock calculations that are responsive to trend and/or seasonality in future forecasts. A forecast tracking signal was used to dampen forecast inaccuracy by adjusting safety stock calculations in cases where forecasts were consistently overoptimistic.

To tackle the unpredictable increase in scrap level Murthy and Ma (1996) developed a mathematical model to calculate the optimal over planning factor required to cater for the increase as a result of supply and process failures. The optimal over planning factor would then be used as a dampening allowance in the subsequent planning process. Through simulation experiments Pandey and Hasin (1998) proposed the use of process lead-time adjustment to tackle scrap and rework in manufacturing due to the difference between planned scrap level and actual scrap level. The process lead-time adjustment technique was merely applied to reduce scrap level, but further exploitation of the technique can be made to tackle other uncertainties.

This review shows that researchers are proposing and practitioners are prepared to provide suitable buffering or dampening techniques to deal with the changes caused by uncertainty in manufacturing enterprises. The existing approach in managing uncertainty prevents further examination on why such uncertainty exists and it does not enable a holistic approach to manage the underlying causes. A holistic approach means being able to dynamically consider the possible uncertainties and their causes and to diagnose the significant underlying causes with reference to the effects on product late delivery, before suggesting suitable buffering or dampening techniques. The review also identifies that little research examined the effects of uncertainty in certain manufacturing environments, i.e. MTS, make-to-order (MTO) and mixed-mode (MM). MTS manufacturing environments can be defined as producing products for stocks and delivering to customer from stock. MTO manufacturing environment can be defined as producing or assembling product for customers' orders. MM manufacturing environment can be defined as a mixture of MTS and MTO with a combination of small and large order intervals and order sizes.

To enable a competitive advantage on delivery to be created and sustained in SMEs that use ERP systems for production planning and control, it is important to understand the underlying principles of how change and uncertainty are managed and tackled; and the effect of uncertainty on product late delivery. This study focused on how ERP could be used to the advantage of SMEs in creating a competitive delivery

performance by managing change and uncertainty using a holistic approach. An uncertainty diagnosing business model (Koh and Saad, 2002, 2003b) was applied for holistic uncertainty management. This study also considered the factor of MTS, MTO and MM manufacturing environments.

Aims and objectives

The aims of this work were to show how ERP could create a competitive advantage for SMEs by being responsive and agile to change and uncertainty, and to identify other methods that are used concurrently for creating a competitive advantage. The objectives of this study were to examine how responsive and agile the existing ERP systems used in SMEs were to change and uncertainty, and to identify the types of change and uncertainty in SME manufacturing environments. The hypothesis was that different manufacturing environments suffer different effects of underlying causes of uncertainty on product late delivery.

Research methodology

The main methodology used in this study was based on the application of an uncertainty diagnosing business model, which structures and relates uncertainty in terms of cause-and-effect (Koh and Saad, 2002). The business model consists of 46 underlying causes of uncertainty and 20 effects, which contributed towards product late delivery. The aim of the business model was to accentuate the underlying causes of uncertainty that significantly affect product late delivery in certain manufacturing environments, so that a manufacturing enterprise could prioritise the attention in finding suitable buffering or dampening techniques to manage the causes that matter.

It was found in Koh *et al.* (2003) that certain types of manufacturing environment would experience similar significant underlying causes of uncertainty. Koh *et al.* (2000b) found that MM manufacturing environment suffers from a greater effect of uncertainty than MTS and MTO environments in enterprises that use ERP systems for production planning and control. Koh *et al.* (2003) found that a wide range of underlying causes of uncertainty significantly affected the level of product late delivery in MTO manufacturing environments. However, application of the business model in SMEs has not been done before. Therefore, the business model was applied in this study for diagnosing the significant underlying causes of uncertainty for SMEs.

The business model was applied in two ways:

- (1) to collect data from SME manufacturers in MTS, MTO and MM environments; and
- (2) to analyse the effects of the underlying causes of uncertainty (structured in the business model) on product late delivery in MTS, MTO and MM manufacturing environments in SMEs.

The data were collected in the form of percentage contributions of the underlying causes of uncertainty (structured in the business model) on product late delivery. The lower the percentages, then the lower the contributions on product late delivery, and vice-versa. Data collection was carried out using a questionnaire administered to 126 UK operations, production, planning or manufacturing managers or directors mainly based in the South of England. The questionnaire was presented in the form of an effect recognition matrix (Koh and Saad, 2003b) for the users to enable valid enquiry

and recognition of the structured cause-and-effect relationships between uncertainties. Some 108 enterprises responded (86 percent response rate) of which 64 were SMEs. SMEs were defined as companies with less than 250 employees but excluded micro enterprises with less than ten employees. Semi-structured telephone interviews were used to verify the responses and the SPSS statistical package was used to analyse the survey results.

Analysis of Variance (ANOVA) was carried out in SPSS to analyse the effects of the underlying causes of uncertainty (structured in the business model) on product late delivery in MTS, MTO and MM manufacturing environments in SMEs. To show that different manufacturing environments will experience different effects of underlying causes of uncertainty on product late delivery, the ANOVA results must show a significant difference ($p < 0.05$) that the underlying causes of uncertainty have an effect on product late delivery.

The business environmental factors in a manufacturing enterprise may affect the product late delivery performance even if the significant underlying causes of uncertainty were managed. Six business environmental factors were also examined (exclude size of enterprise since this study is focused on SMEs) and the results were analysed using descriptive analyses.

Results, analysis and discussions

Results from the one-way ANOVA showed that a different group of underlying causes of uncertainty significantly affects the product late delivery performance in MTS, MTO and MM manufacturing environments in SMEs. Since, these underlying causes of uncertainty and their effects could occur at a different time in a manufacturing environment, one-way ANOVA is conducted to give an overview of which underlying causes of uncertainty were found to be more significant than the others.

Significant uncertainty in MTS environments

Table I summarises the significant ANOVA results for MTS manufacturing environment in SMEs. In the MTS manufacturing environment, a total of seven underlying causes of uncertainty were found to be significantly affecting product late delivery in these SMEs. These were insecure stores, customer changes delivery lead-times, internal design changes during/after planning, MRP plan overload (infinite scheduling of labour), planned maintenance/repair time exceeded, waiting for labour, and seeking concession.

It was envisaged that insecure stores would not result in such an impact on delivery due to the reliance of product delivery from stores in an MTS environment. The effect from “customer changes delivery lead-times” would be dependent on the magnitude of the change from the original lead-times. The shorter the new delivery lead-time, the larger would be the effect on product late delivery. Owing to the possibility of product shortages from stores for delivery, a new order might be released for production and thus could result in late delivery.

It was a surprise to find that internal design changes during/after planning significantly affected product late delivery in MTS environments. Since, this uncertainty would occur at the planning stage, changes could still be made and thus minimise the effect on delivery. Unless the design changes involved a major redesign within which the time expected is exceeded, some product shortages might occur and some orders might be delivered late. MRP plan overload (infinite scheduling of labour)

Significant underlying causes of uncertainty		Sum of squares	df	Mean square	<i>F</i>	Sig. (<i>p</i>)	Competitive advantage for small businesses
Insecure stores	Between groups	3975.150	7	567.879	4.968	0.008**	
	Within groups	1371.800	12	114.317			
	Total	5346.950	19				
Customer changes delivery lead-times	Between groups	5887.000	7	841.000	3.212	0.037*	
	Within groups	3141.950	12	261.829			
	Total	9028.950	19				
Internal design changes during/after planning	Between groups	3267.850	7	466.836	11.920	0.000**	
	Within groups	469.950	12	39.163			
	Total	3737.800	19				
MRP plan overload (infinite scheduling of labour)	Between groups	8281.550	7	1183.079	4.772	0.009**	
	Within groups	2975.000	12	247.917			
	Total	11256.550	19				
Planned maintenance/repair time exceeded	Between groups	4678.950	7	668.421	7.086	0.002**	
	Within groups	1132.000	12	94.333			
	Total	5810.950	19				
Waiting for labour	Between groups	13852.600	7	1978.943	10.691	0.000**	
	Within groups	2221.200	12	185.100			
	Total	16073.800	19				
Seeking concessions	Between groups	10608.000	7	1515.429	32.312	0.000**	
	Within groups	562.800	12	46.900			
	Total	11170.800	19				

Notes: **p* < 0.05; ***p* < 0.01

Table I.
ANOVA results of significant underlying causes of uncertainty on product late delivery for MTS manufacturing environment in SMEs

was expected to result in product late delivery due to the required resources unavailability. Even if product could be delivered from stores, the rate to fill in the stock would be compromised. Planned maintenance/repair time exceeded and waiting for labour would have such similar effects. Seeking concession happens at the post-production stage; hence its occurrence would directly affect product late delivery, but not on resources loading and allocation.

Significant uncertainty in MTO environments

Table II shows the significant ANOVA results for MTO manufacturing environments in SMEs. In the MTO manufacturing environment, a total of 30 underlying causes of uncertainty were found to be significantly affecting product late delivery in these SMEs. The range of significant underlying causes of uncertainty identified in MTO environment is larger than those in MTS environment because of the relatively dynamic and stochastic nature of MTO environment. Problems related to supplier delivery performance, customer or demand changes, resources (labour, machine and material) availability, design changes, schedule control rules, quality issues, and post-production are all significantly affecting product late delivery in the MTO environment. Given that the results from interviewing these SMEs show that they use various buffering or dampening techniques to tackle uncertainty, it appears that the effectiveness and efficiency of those techniques were questionable considering the significance of the underlying causes of uncertainty found in such an environment.

Significant underlying causes of uncertainty		Sum of squares	df	Mean square	<i>F</i>	Sig. (<i>p</i>)
Rejected by quality	Between groups	8705.392	6	1450.899	8.577	0.002**
	Within groups	1691.667	10	169.167		
	Total	10397.059	16			
Supplier delivered with shortages	Between groups	7433.059	6	1238.843	25.283	0.000**
	Within groups	490.000	10	49.000		
	Total	7923.059	16			
Supplier late delivery	Between groups	17305.275	6	2884.212	195.319	0.000**
	Within groups	147.667	10	14.767		
	Total	17452.941	16			
Incorrect items supplied	Between groups	265.569	6	44.261	3.794	0.031*
	Within groups	116.667	10	11.667		
	Total	382.235	16			
Items missing in BOM	Between groups	1985.804	6	330.967	160.145	0.000**
	Within groups	20.667	10	2.067		
	Total	2006.471	16			
Unexpected demand pattern changes	Between groups	14375.882	6	2395.980	11.041	0.001**
	Within groups	2170.000	10	217.000		
	Total	16545.882	16			
Demand/usage analysis not used to drive stock control rules	Between groups	6297.451	6	1049.575	8.487	0.002**
	Within groups	1236.667	10	123.667		
	Total	7534.118	16			
Customer changes delivery lead-times	Between groups	4245.098	6	707.516	8.663	0.002**
	Within groups	816.667	10	81.667		
	Total	5061.765	16			
Customer changes product ordered	Between groups	4126.471	6	687.745	12.504	0.000**
	Within groups	550.000	10	55.000		
	Total	4676.471	16			
Customer changes specified level of quality	Between groups	171.569	6	28.595	17.157	0.000**
	Within groups	16.667	10	1.667		
	Total	188.235	16			
Customer design changes during/after planning	Between groups	1471.569	6	245.261	9.197	0.001**
	Within groups	266.667	10	26.667		
	Total	1738.235	16			
Internal design changes during/after planning	Between groups	271.098	6	45.183	21.863	0.000**
	Within groups	20.667	10	2.067		
	Total	291.765	16			
Sickness	Between groups	8967.971	6	1494.662	16.607	0.000**
	Within groups	900.000	10	90.000		
	Total	9867.971	16			
Holiday	Between groups	12827.618	6	2137.936	23.755	0.000**
	Within groups	900.000	10	90.000		
	Total	13727.618	16			
Schedule/work-to-list not produced	Between groups	21811.059	6	3635.176	22.357	0.000**
	Within groups	1626.000	10	162.600		
	Total	23437.059	16			
Schedule/work-to-list not controlled	Between groups	15295.216	6	2549.203	13.686	0.000**
	Within groups	1862.667	10	186.267		
	Total	17157.882	16			
Schedule/work-to-list produced but not available to labour	Between groups	2515.216	6	419.203	251.522	0.000**
	Within groups	16.667	10	1.667		

(continued)

Table II.
ANOVA results of significant underlying causes of uncertainty on product late delivery for MTO manufacturing environment in SMEs

Significant underlying causes of uncertainty		Sum of squares	df	Mean square	F	Sig. (p)
MRP plan overload (infinite scheduling of labour)	Total	2531.882	16			
	Between groups	16545.098	6	2757.516	13.343	0.000 **
	Within groups	2066.667	10	206.667		
Unexpected/urgent changes to schedule (labour assignment)	Total	18611.765	16			
	Between groups	16545.098	6	2757.516	13.343	0.000 **
	Within groups	2066.667	10	206.667		
Planned maintenance/repair time exceeded	Total	18611.765	16			
	Between groups	6827.451	6	1137.908	5.643	0.009 **
	Within groups	2016.667	10	201.667		
Planned set-up/change-over time exceeded	Total	8844.118	16			
	Between groups	18273.529	6	3045.588	203.039	0.000 **
	Within groups	150.000	10	15.000		
MRP plan overload (infinite scheduling of machine)	Total	18423.529	16			
	Between groups	15021.451	6	2503.575	44.654	0.000 **
	Within groups	560.667	10	56.067		
Waiting for internal material	Total	15582.118	16			
	Between groups	9959.804	6	1659.967	15.562	0.000 **
	Within groups	1066.667	10	106.667		
Waiting for external material	Total	11026.471	16			
	Between groups	11256.863	6	1876.144	45.027	0.000 **
	Within groups	416.667	10	41.667		
Labour error	Total	11673.529	16			
	Between groups	9511.098	6	1585.183	6.831	0.004 **
	Within groups	2320.667	10	232.067		
Defective raw material	Total	11831.765	16			
	Between groups	11885.804	6	1980.967	958.533	0.000 **
	Within groups	20.667	10	2.067		
Customer design changes during/after production	Total	11906.471	16			
	Between groups	28537.235	6	4756.206	79.138	0.000 **
	Within groups	601.000	10	60.100		
Items on hold (financial)	Total	29138.235	16			
	Between groups	25970.098	6	4328.350	472.184	0.000 **
	Within groups	91.667	10	9.167		
Unavailability of transport	Total	26061.765	16			
	Between groups	21992.569	6	3665.428	7.533	0.003 **
	Within groups	4865.667	10	486.567		
Seeking concession	Total	26858.235	16			
	Between groups	463.059	6	77.176	5.011	0.013 *
	Within groups	154.000	10	15.400		
	Total	617.059	16			

Notes: * $p < 0.05$; ** $p < 0.01$

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Table II.

In the MTO environment, supplier delivery under-performance in terms of late delivery, rejection rate, shortages and delivered items out of specifications, were found to be significantly affecting product late delivery. This finding shows that integration of supply chain management and supplier network management with customer order management and production planning and control in SMEs had been overlooked. The ANOVA results also show that the BOM was not updated accordingly and hence

resulted in items missing in BOM when generating a POR schedule. Product late delivery was also found to be caused by the lack of customer demand analysis. Owing to the dynamic and stochastic nature of MTO environments, changes made by customers were found to overwhelmingly cause product late delivery. It appeared that there was a lack of a procedure or policy to handle customer changes. Although changes to customer order can be managed in an ERP system, such changes are not reflected in the system, as the POR schedule has already been released, i.e. late changes. This finding showed that early changes during planning can be managed by an ERP system before POR schedule is released, otherwise it was too late to update the plan and subsequently an alternative method needed to be devised to tackle the uncertainty.

It was expected that design changes would cause product late delivery in MTO environments as orders are usually treated as a form of contract and a variety of standardised and customised orders and requests are handled. Labour unavailability in terms of sickness, holiday, MRP plan overload, and unexpected/urgent changes to schedule were found to be a major cause of product late delivery. The results verification process via interviewing the SMEs found that they were surprised with this finding and were not aware of that causing overall late delivery. This might be due to the more flexible nature in these SMEs, which did not holistically manage change and uncertainty. Machine unavailability in terms of planned maintenance/repair time exceeded, planned set-up/change-over time exceeded, MRP plan overload, and waiting for internal and external materials were also found to significantly cause product late delivery due to halting of resources for production.

Schedule/work-to-list was not controlled and not efficiently handled in the shop floor, thus they were found significantly affecting delivery performance. Scrap/rework due to labour error and defective raw material were also determinants of late delivery in MTO environments. This might be due to the process and material changes required to respond to customer product and specification changes. Since, there were many changes and uncertainties in MTO environments, it was not unusual to find that even after production was completed, some post-production uncertainties did cause late delivery, i.e. items on hold due to financial clearance, unavailability of transport due to logistics problems, and seeking concession.

Significant uncertainty in MM environment

Table III summarises the significant ANOVA results for MM manufacturing environment in SMEs. In the MM manufacturing environment, a total of five underlying causes of uncertainty were found to be significantly affecting product late delivery in these SMEs. These are insecure stores, customer changes delivery lead-times, MRP plan overload (infinite scheduling of labour), waiting for labour, and seeking concession. The range of significant underlying causes of uncertainty identified in MM environment was relatively less than those in MTO environments. This finding could be described as the phenomenon of the stabilising effect from the MTS element in MM environment, which has removed a vast amount of possible significant uncertainties.

The nature of MM environments, i.e. combination of MTS and MTO, has resulted in the finding that insecure stores would cause product late delivery due to the element of the shortages of delivery from stores. Customer changes delivery

Significant underlying causes of uncertainty		Sum of squares	df	Mean square	F	Sig. (p)
Insecure stores	Between groups	5001.467	12	416.789	2.583	0.47 *
	Within groups	2259.200	14	161.371		
	Total	7260.667	26			
Customer changes delivery lead-times	Between groups	6633.033	12	552.753	2.907	0.030 *
	Within groups	2661.633	14	190.117		
	Total	9294.667	26			
MRP plan overload (infinite scheduling of labour)	Between groups	20983.333	12	1748.611	6.177	0.01 **
	Within groups	3963.333	14	283.095		
	Total	24946.667	26			
Waiting for labour	Between groups	17157.374	12	1429.781	4.261	0.06 **
	Within groups	4698.033	14	335.574		
	Total	21855.407	26			
Seeking concession	Between groups	4418.533	12	368.211	7.060	0.000 **
	Within groups	730.133	14	52.152		
	Total	5148.667	26			

Notes: * $p < 0.05$; ** $p < 0.01$

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Table III.
ANOVA results of significant underlying causes of uncertainty on product late delivery for MM manufacturing environment in SMEs

lead-times might require the existing order to be fulfilled partially through stock or a revised order might be released for production. Insufficient product in stores could result in late delivery, whilst a new order would require new entry into the order queue thus affecting resources allocation and might result in late delivery.

MRP plan overload (infinite scheduling of labour) was expected to result in product late delivery due to the required resources unavailability. Even if some parts could be delivered from stores, those that need to be ordered and produced would still be delayed. Waiting for labour had the same effect likewise. Seeking concession was also found to be causing late delivery, but this underlying cause was found to be significant in all MTS, MTO and MM environments. Therefore, regardless of the manufacturing environment, seeking concession is an issue in the post-production stage that all SMEs should be aware of.

Significant uncertainty – a summary

The overall results suggested that different types of underlying causes of uncertainty significantly affected the product late delivery in different manufacturing environment in SMEs. The results also showed that the effects and the ranges of underlying causes of uncertainty in MTO manufacturing environments appear to be significantly higher than in MTS and MM environments. This finding was contradicted in Koh *et al.* (2000b), which found that MM environment is significantly affected by uncertainty. However, there is a major difference between these studies in that the sample used by Koh *et al.* (2000b) was inclusive of all sizes of enterprises but this study only looked at SMEs. This could lead to the finding that larger MM manufacturing enterprises that use ERP for production planning and control may be affected by a larger range of significant uncertainties.

This study showed that the results from ANOVA were be useful for SMEs in MTS, MTO and MM manufacturing environments in providing a guideline to prioritise the effort in devising suitable buffering or dampening techniques to manage uncertainty.

This approach of managing uncertainty will be more holistic because the possible ranges of underlying causes of uncertainty that significantly affect product late delivery could be considered simultaneously whilst deciding upon the suitable buffering or dampening techniques in a specific condition.

Business environmental factors

Tables IV–IX show the results from descriptive analyses on the degree of influence of the business environmental factors in MTS, MTO and MM manufacturing environments in SMEs. The number of respondents (total $n = 64$) in MTS, MTO and MM environments were found to be 20, 17 and 27 UK SMEs, respectively. The results clearly showed that product late delivery performance in medium-sized MTO and MM manufacturing enterprises were mainly affected by the significant underlying causes of uncertainty found from the use of the business model. However, a similar proportion of SMEs were found to be effective in using the significant underlying causes of uncertainty diagnosed for MTS environments. The type of manufacturing sectors that an enterprise is competing in was found to be not influential in determining the effects of uncertainty due to the mix and spread of sectors.

The results showed that MTS, MTO and MM manufacturing environments with less than 5,000 parts mix were likely to be influenced by the effects of the significant underlying causes of uncertainty found in these environments. The results evidenced

	Size of enterprises (UK DTI definition)		
Small	11 (55)	6 (35)	9 (33)
Medium	9 (45)	11 (65)	18 (67)
Influential ranges	MTS	MTO	MM
<i>1. Manufacturing sector</i>			
Basic metals	0 (0)	2 (11.7)	0 (0)
Chemicals, chemical products and man-made fibres	1 (5)	1 (5.9)	3 (11.1)
Electrical machinery and apparatus not elsewhere classified	3 (15)	2 (11.7)	3 (11.1)
Fabricated metal products, except machinery and equipment	4 (20)	2 (11.7)	4 (14.8)
Food products, beverages and tobacco	0 (0)	0 (0)	1 (3.7)
Machinery and equipment not else where classified	4 (20)	2 (11.7)	5 (18.5)
Manufacturing not elsewhere classified	0 (0)	0 (0)	1 (3.7)
Medical precision and optical instruments, watches and clocks	3 (15)	0 (0)	3 (11.1)
Motor vehicles, trailers and semi-trailers	0 (0)	2 (11.7)	1 (3.7)
Office machinery and computers	0 (0)	0 (0)	1 (3.7)
Other non-metallic mineral products	2 (10)	0 (0)	2 (7.4)
Other transport equipment	0 (0)	2 (11.7)	0 (0)
Publishing, printing and reproduction of recorded media	0 (0)	2 (11.7)	0 (0)
Radio, television and communication equipment and apparatus	1 (5)	2 (11.7)	1 (3.7)
Rubber and plastic products	2 (10)	0 (0)	2 (7.4)

Table IV.
The influence of manufacturing sector (according to Office of National Statistics UK, 1998) in MTS, MTO and MM manufacturing environments in SMEs ($n = 64$)

Note: Figures given in parentheses are percentages

that MTS, MTO and MM manufacturing enterprises that use more than 50 percent of bought-out parts in production are likely to be affected. Despite this finding, many other underlying causes of uncertainty (as opposed to external supply) were found to be significant. Therefore, a large buy versus make ratio will not guarantee a manufacturing environment that will be affected by less uncertainty.

The majority of the SMEs were found to abandon the use of a rough-cut capacity plan. Nonetheless, a slightly larger number of MTS and MM enterprises used this type of capacity planning. These findings showed that the use of rough-cut capacity planning influences the effects of the underlying causes of uncertainty on product late delivery in MTS and MM manufacturing environments given that plan overload was found to be a major problem in these environments. The results showed major

	Size of enterprises (UK DTI definition)		
Small	11 (55)	6 (35)	9 (33)
Medium	9 (45)	11 (65)	18 (67)
Influential ranges	MTS	MTO	MM
<i>2. Number of parts</i>			
<1,000	9 (45)	8 (47)	14 (52)
>1,000 and <5,000	10 (50)	9 (53)	11 (41)
>5,000	1 (5)	0 (0)	2 (7)

Note: Figures given in parentheses are percentages

Table V.
The influence of number
of parts in MTS, MTO
and MM manufacturing
environments in SMEs
(*n* = 64)

	Size of enterprises (UK DTI definition)		
Small	11 (55)	6 (35)	9 (33)
Medium	9 (45)	11 (65)	18 (67)
Influential ranges	MTS	MTO	MM
<i>3. Ratio of buy versus make parts</i>			
Approximately 60:40	9 (45)	11 (65)	10 (37)
Approximately 50:50	6 (30)	6 (35)	12 (44)
Approximately 40:60	3 (15)	0 (0)	3 (11)
Approximately 20:80	2 (10)	0 (0)	2 (8)

Note: Figures given in parentheses are percentages

Table VI.
The influence of the ratio
of buy versus make parts
in MTS, MTO and MM
manufacturing
environments in SMEs
(*n* = 64)

	Size of enterprises (UK DTI definition)		
Small	11 (55)	6 (35)	9 (33)
Medium	9 (45)	11 (65)	18 (67)
Influential ranges	MTS	MTO	MM
<i>4. Rough-cut capacity planning</i>			
No	12 (60)	14 (82)	15 (56)
Yes	8 (40)	3 (18)	12 (44)

Note: Figures given in parentheses are percentages

Table VII.
The influence of
rough-cut capacity
planning in MTS, MTO
and MM manufacturing
environments in SMEs
(*n* = 64)

underperformance of the buffering or dampening techniques used by the SMEs in managing uncertainty and a few suggested that the use of these techniques could be further improved.

The type of changes that could be responded to using an ERP system was examined through the survey and interview. We found from the survey that the majority of SMEs only utilise the feature for changing dimension for product design and that there was no difference between SMEs that operate in MTS, MTO and MM environments. Only a few SMEs in MTO and MM environments extended the due date of customer order using their ERP systems, and none were found using the ERP system to shorten supply lead-times. Further, verification of the survey results through interviews suggested that the SMEs mainly used the ERP system to generate a plan for production and using it as a guideline. There is a lack of knowledge and expertise in how further modification, feedback and management can be made and arranged to enable their ERP systems to perform better as a production planning and control tool. There was also an absence of tools and techniques to strategically manage uncertainty and change.

ERP in SMEs

The main finding of this study was that ERP could improve responsiveness and agility to change, but not to uncertainty. Therefore, SMEs could create a competitive advantage by being more responsive to change through the use of ERP as a production planning and control tool. These changes refer to any information update necessary, for example, due date extension of customer order, dimension changes in product design, and shortening supply lead-times, in the ERP system before generating purchase and work order. However, the changes are the effects from uncertainty. ERP

Table VIII.
The influence of the extent of the use buffering and dampening techniques in production in MTS, MTO and MM manufacturing environments in SMEs (*n* = 64)

	Size of enterprises (UK DTI definition)		
	Small	Medium	Influential ranges
	11 (55)	9 (45)	MTS
	6 (35)	11 (65)	MTO
<i>5. Extent of the use buffering and dampening techniques in production</i>			
Used but not satisfied	12 (60)	9 (53)	18 (67)
Used but satisfied (could be further improved)	8 (40)	8 (47)	9 (33)
Note: Figures given in parentheses are percentages			

Table IX.
The influence of changes responded to using the ERP system in MTS, MTO and MM manufacturing environments in SMEs (*n* = 64)

	Size of enterprises (UK DTI definition)		
	Small	Medium	Influential ranges
	11 (55)	9 (45)	MTS
	6 (35)	11 (65)	MTO
<i>6. Changes responded to using the ERP system</i>			
Due date extension of customer order	0 (0)	5 (29)	3 (11)
Dimension changes in product design	14 (70)	10 (59)	18 (67)
Shortening supply lead-times	0 (0)	0 (0)	0 (0)
Note: Figures given in parentheses are percentages			

systems could not deal with uncertainty due to its stochastic and unpredictable nature, which often occurred after purchase and work order were issued and released. As a result, many SMEs used a range of buffering or dampening techniques under uncertainty to be competitive in delivery.

ERP could create a competitive advantage in terms of delivery performance in SMEs by holistically managing underlying causes of uncertainties that significantly affect product late delivery in their specific manufacturing environments, i.e. MTS, MTO or MM. Different types of uncertainty affected SMEs in MTS, MTO and MM manufacturing environments, respectively. It is important for SMEs in each manufacturing environment to recognise the significant underlying causes and tackle them urgently using appropriate buffering or dampening techniques. The findings on the significant underlying causes in this study can be used as a knowledge base for the SMEs in each manufacturing environment for uncertainty management. Since, these findings were derived using a specific sample type, it is also crucial to recognise the business environmental factors considered in this study to find the condition which best suits the knowledge base. When the significant underlying causes of uncertainty are tackled holistically, the required changes in ERP systems would be reduced, and the product delivery performance could be improved.

Conclusions and implications

The main conclusion from this study was that ERP could create a competitive advantage in delivery for SMEs by being responsive and agile to change, but not to uncertainty. The survey and interview results suggested that only a few features in an ERP system (i.e. mainly dimension changes for product design) were used to deal with change due to uncertainty. It was found that SMEs generally use their ERP system to generate a plan for production and use it as a guideline. This study revealed that the SMEs concurrently use a range of buffering or dampening techniques to tackle uncertainty for creating a competitive advantage in delivery. It was also identified that there was a lack of knowledge and expertise in how further modification, feedback and management can be made and arranged to enable their ERP systems to perform better as a production planning and control tool. An absence of tools and techniques to strategically manage uncertainty and change in SMEs was also uncovered.

In this study, an uncertainty diagnosis business model for uncertainty management was applied in SMEs. A questionnaire survey was carried out with UK manufacturing enterprises that use ERP systems for production planning and control to collect data in order to prove the hypothesis that product late delivery is significantly affected by different underlying causes of uncertainty in different manufacturing environments (i.e. MTS, MTO and MM). The ANOVA results showed some 30 underlying causes of uncertainty significantly affect product late delivery in a MTO environment. Seven underlying causes of uncertainty were found to significantly affect product late delivery in MTS environments. While in MM environments only five underlying causes of uncertainty were found to significantly affect product late delivery. The ANOVA results confidently proved the hypothesis, and it can be concluded that different manufacturing environments suffer different effects of underlying causes of uncertainty on product late delivery. The product late delivery performance in MTO manufacturing environments was found to be significantly affected by a wide range of underlying causes of uncertainty. It was interesting to find that MM manufacturing

environments had an opposite outcome. This could be due to the effective use of buffering or dampening techniques in the sample used. Further analysis will be made when investigating the types of techniques used and how they were applied in such an environment. Further research will also be carried out to study what buffering or dampening techniques are the most robust in tackling uncertainty in MTS and MTO SMEs manufacturing environments.

The SMEs' business environmental factors were analysed to give further insights into the characteristics of the SMEs examined. It can be concluded from the descriptive analyses of the business environmental factors for MTS, MTO and MM environments that number of parts, ratio of buy versus make parts, the use of rough-cut capacity planning, the use of buffering or dampening techniques in production and the type of changes responded using an ERP system influenced the effects of underlying causes of uncertainty on product late delivery. Considering these factors, it can be concluded that the significant underlying causes of uncertainty can be used as a knowledge base for the specific SMEs in each manufacturing environment for uncertainty management. Researchers and practitioners could use these findings to prioritise their efforts in managing and tackling uncertainties. When the significant underlying causes of uncertainty are tackled holistically, the required changes in ERP systems would be reduced, and the product delivery performance could be improved.

A holistic approach for managing uncertainty was required (Koh and Saad, 2003b) and hence we proposed the use of the business model for uncertainty management. It can be concluded that the application of the business model in SMEs has provided useful knowledge to MTS, MTO and MM manufacturing enterprises on which underlying causes of uncertainty were significantly affecting their product late delivery performance. Using this knowledge, SMEs could then prioritise the effort and devise suitable buffering or dampening techniques. This research showed that a holistic approach such as the business model gives a solid foundation for enterprises to evaluate their performance.

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